

HIGH PERFORMANCE QUADCOPTER

System Requirements Document

Rocket Geometry Airframe | 6S Build | Iteration 1 of 2

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1. Purpose

This document defines the system-level requirements for Iteration 1 of the High Performance Quadcopter project. This iteration uses a 6S power system with the AOS Supernova 2207 1980KV motor to validate the custom airframe design, establish a performance baseline, and inform the design decisions for Iteration 2 which targets 200+ km/h on an 8S power system.

All design decisions made throughout the project shall be traceable to the requirements defined here. This project develops practical aerospace engineering skills including aerodynamic shaping, structural design, propulsion sizing, CFD analysis, and flight test data analysis.

2. Mission Overview

2.1 Mission Statement

Design, fabricate, and flight-test a high performance quadcopter featuring a custom streamlined rocket-geometry fuselage, fabricated from carbon fiber filled filament, powered by a 6S electrical system with AOS Supernova 2207 1980KV motors and 5-inch propellers. Target top speed is 160-175 km/h in calm air, with 185-200 km/h achievable downwind.

2.2 Iteration Strategy

This project is structured as two deliberate engineering iterations:

- Iteration 1 (this document): 6S power system — validate airframe, establish performance baseline, document lessons learned
- Iteration 2 (future): 8S power system with AOS Supernova 1570KV — target 200+ km/h in calm air based on Iteration 1 data

This two-iteration approach is intentional engineering practice. Iteration 1 data directly informs Iteration 2 design decisions. Both iterations together form a stronger portfolio narrative than a single build.

2.3 Concept of Operations

The vehicle operates outdoors in open areas under direct pilot RC control with visual line of sight maintained at all times. Flight operations progress through structured phases:

- Phase 1: Hover and stability validation — verify airframe integrity and establish PID baseline
- Phase 2: Progressive speed buildup — incremental speed runs with data logging at each stage
- Phase 3: Maximum speed runs — GPS-logged runs in calm air and downwind to measure top speed
- Phase 4: Characterization — document flight time, handling, and full performance envelope

3. Performance Requirements

Note: Speed targets are for Iteration 1 (6S). Iteration 2 (8S) will target 200+ km/h in calm air.

ID	Requirement	Priority
PR-01	Vehicle shall achieve a minimum top speed of 140 km/h in calm air conditions	High
PR-02	Vehicle shall achieve a target top speed of 160-175 km/h in calm air conditions	High
PR-03	Vehicle shall achieve 185-200 km/h ground speed in favorable downwind conditions (20-30 km/h tailwind)	Medium
PR-04	Vehicle thrust-to-weight ratio shall exceed 10:1 at full throttle on 6S	High
PR-05	Vehicle shall achieve a minimum flight time of 3 minutes at full throttle	High
PR-06	Vehicle shall achieve a minimum flight time of 5 minutes at 60% throttle cruise	Medium
PR-07	Vehicle shall maintain stable controlled flight at speeds up to 120 km/h minimum	High
PR-08	Vehicle shall achieve stable hover without pilot correction for a minimum of 5 seconds	High
PR-09	Theoretical top speed shall be calculated from motor KV, voltage, and prop pitch before first flight and compared to GPS measured speed in test report	High

4. Airframe & Structural Requirements

ID	Requirement	Priority
AF-01	Frame shall be designed entirely from scratch in Fusion 360 — no templates or downloaded STL files permitted	High
AF-02	Frame shall be fabricated from carbon fiber filled filament (PA-CF, PLA-CF, or equivalent)	High
AF-03	Center body shall incorporate a streamlined fuselage geometry to reduce frontal drag versus a standard flat plate center stack	High

ID	Requirement	Priority
AF-04	Fuselage drag reduction shall be quantified by CFD analysis in SimScale and documented in the project report	High
AF-05	Frame arms shall be designed with wall thickness and geometry calculated for aerodynamic and motor torque loads at maximum speed	High
AF-06	All printed parts shall be oriented during slicing to align layer lines with primary load direction	High
AF-07	Vehicle center of gravity shall be within 5mm of geometric center in X and Y axes	High
AF-08	Frame shall be compatible with standard 5-inch prop motor spacing (M3 mounting, 16x16mm or 19x19mm pattern)	High
AF-09	Arms shall be designed as replaceable components to allow crash repair without reprinting the full frame	Medium
AF-10	Frame mass budget shall be estimated at design phase — weight of each printed component recorded before printing	Medium

5. Propulsion System Requirements

ID	Requirement	Priority
PS-01	Motors shall be AOS Supernova 2207 1980KV — rated for 5-6S operation	High
PS-02	ESCs shall be rated for a minimum of 35A continuous current on 6S (25.2V)	High
PS-03	Propellers shall be HQ 5x4.5x3 V1S — the reference tested prop for this motor	High
PS-04	Battery shall be 6S LiPo, 1000-1300mAh capacity, minimum 100C discharge rating	High
PS-05	Motor selection shall be justified by documented thrust analysis showing predicted thrust at full throttle on 6S	High
PS-06	Propeller selection shall be justified by pitch speed calculation showing theoretical maximum speed	High
PS-07	Total system weight including battery shall be calculated at design phase to verify thrust-to-weight requirement PR-04	High
PS-08	Design reference thrust value shall be 1,980g per motor at full throttle on 6S per published test data	High

6. Electronics Requirements

ID	Requirement	Priority
EL-01	Flight controller shall be SpeedyBee F7 V3 or equivalent — F7 processor required for high speed control loop demands	High
EL-02	RC receiver shall be compatible with existing transmitter and configured with failsafe	High

ID	Requirement	Priority
EL-03	All electronics shall be mounted with vibration isolation to protect IMU accuracy at high speed	High
EL-04	Flight controller shall log IMU data, motor outputs, and GPS speed to onboard blackbox every flight	High
EL-05	Wiring shall be routed and secured inside or flush with fuselage to minimize aerodynamic interference	Medium

7. Documentation Requirements

ID	Requirement	Priority
DOC-01	All design decisions shall be documented with engineering rationale at time of decision — not reconstructed after the fact	High
DOC-02	Theoretical top speed calculation shall be committed to GitHub before first flight	High
DOC-03	Thrust-to-weight calculation using actual component weights shall be committed to GitHub before first flight	High
DOC-04	Formal test procedures shall be written prior to executing any test	High
DOC-05	All flight test sessions shall produce a written test report comparing actual results to predictions	High
DOC-06	All files shall be committed to GitHub throughout the project with descriptive commit messages — not uploaded at completion	High
DOC-07	An Iteration 1 lessons learned report shall be produced at completion to formally inform Iteration 2 design	High
DOC-08	Final project report shall be produced in LaTeX and published to the repository	Medium

8. Budget — Iteration 1 (6S)

Component	Est. Cost	Actual Cost	Notes
AOS Supernova 2207 1980KV Motors x4	\$60 - \$80		Best 6S thrust in class
ESCs x4 (35A+ 6S BLHeli32)	\$40 - \$60		Aikon AK32 or Hobbywing XRotor
SpeedyBee F7 V3 Flight Controller	\$40 - \$55		F7 processor, 6S compatible
HQ 5x4.5x3 V1S Propellers + spares	\$15 - \$20		Reference prop for this motor

Component	Est. Cost	Actual Cost	Notes
6S LiPo Battery 1000-1300mAh 100C+	\$40 - \$55		Buy 2 if budget allows
6S capable balance charger	\$25 - \$40		ISDT Q6 Plus recommended
RC Receiver	\$15 - \$25		Must match existing transmitter
Carbon fiber filled filament 1kg	\$30 - \$50		PA-CF or PLA-CF
Miscellaneous (XT60, wiring, hardware)	\$15 - \$20		
Iteration 1 Total	\$280 - \$405		Iteration 2 adds ~\$80-120

9. Stretch Goals

The following are not required for Iteration 1 success but are targeted for future phases.

ID	Requirement	Priority
SG-01	FPV camera and video transmitter for first person view and speed run footage	Low
SG-02	GPS module for precise speed measurement during speed runs	Low
SG-03	Iteration 2: Upgrade to 8S power system (AOS Supernova 1570KV) targeting 200+ km/h in calm air	Low
SG-04	Extended CFD analysis comparing multiple fuselage geometries to optimize drag reduction	Low

10. Success Criteria — Iteration 1

Iteration 1 is considered successfully complete when all of the following are demonstrated:

- Vehicle achieves stable controlled hover for a minimum of 30 seconds
- Vehicle achieves a GPS-measured top speed exceeding 140 km/h
- Vehicle achieves minimum 3 minutes flight time at full throttle
- Theoretical top speed calculation produced and compared to measured result in test report
- Thrust-to-weight ratio verified to exceed 10:1 by calculation using actual component weights
- All High priority requirements verified through formal test procedures
- CFD analysis of fuselage produced in SimScale and documented in report
- Lessons learned report produced to formally inform Iteration 2 design
- Complete documentation published to GitHub with timestamped commit history
- Frame is an original Fusion 360 design — not a template or downloaded file

11. Revision History

Rev	Date	Description
A	2/16/2026	Initial release — 4S 5 inch configuration
B	2/17/2026	Revised to 6S high performance, added rocket geometry fuselage requirements
C	2/19/2026	Revised to 6S 5 inch — Iteration 1 of 2. Iteration 2 targets 8S 200+ km/h. Added SimScale CFD requirement.